

Assessing the Energy Potential of Artificial Intelligence Data Center Sites

A Framework for Comparing Site Suitability

Artificial intelligence (AI) development is rapidly increasing electricity demand in the United States. Developers and operators of large AI data centers generally prefer grid connections because they offer greater reliability than on-site power generation alone. The challenge is no longer only how much additional power the United States might add by 2030: It is also **where that power can realistically be delivered**. For frontier AI data centers, site selection will increasingly depend on existing energy infrastructure, local system conditions, and other place-specific constraints.

To help address this challenge, RAND researchers examined two questions: (1) Which sites or regions have favorable or unfavorable characteristics for maximizing energy capacity for AI data centers, and (2) what is the maximum amount of electric

KEY FINDINGS

- RAND researchers developed a **five-part framework** for evaluating the energy-related suitability of potential artificial intelligence (AI) data center sites.
- **Existing infrastructure**—especially substations, transmission access, and previously developed industrial sites—provides a major advantage for delivering power by 2030.
- **No site is ideal across all dimensions**; each involves trade-offs among power availability, infrastructure readiness, environmental risk, and governance factors.
- Among the sites examined, a small number stood out as having relatively high potential to be AI data center sites.
- The researchers assessed three retired or retiring power plant sites to estimate how much capacity could realistically be made available at one location by 2030. Of these, **Rockport Power Plant in Indiana** had the highest estimated potential, at about **4.2 gigawatts (GW)**.
- Given the difficulty of building major new transmission and distribution infrastructure before 2030, the Rockport estimate of about 4.2 GW may represent the **practical upper bound of grid-connected power available at a single U.S. site** on that timeline.

power that the grid can deliver at a single site by 2030? To answer these questions, the researchers developed a framework for comparing site energy suitability and applied it to 22 locations: 17 sites identified by the U.S. Department of Energy (DOE), two major private-sector sites, and three retired or retiring power plant sites. The framework is intended to serve as an energy-potential screening tool for planners, policymakers, and developers.

The Framework

The framework assesses sites across five categories that together shape how favorable a location is, from an energy perspective, for AI data center development. These categories are intended to capture whether energy can likely be delivered reliably and at scale by 2030 at the site. Because this analysis focuses on site suitability from an energy perspective, the framework emphasizes factors that affect whether power can be supplied, delivered, and supported at scale. Table 1 summarizes the five categories used to assess those energy-related conditions across candidate sites.

These categories are grounded in a broad conceptual view of how site characteristics influence available energy capacity. Figure 1 shows how infrastructure, enabling factors, and barriers combine to shape the five categories used in the assessment.

Some Sites Stand Out, but No Site Is Perfect

The researchers did not identify one universally best site. Instead, sites fell into broad tiers of **high**, **moderate**, and **low** energy potential.

High-potential sites tended to combine stronger existing infrastructure, favorable supporting inputs, and fewer major bottlenecks. Examples included **Stargate**, the **Pantex Plant**, and the **Kansas City National Security Campus**. These locations generally benefited from some combination of existing transmission access, available land, favorable regulatory conditions, and nearby energy resources.

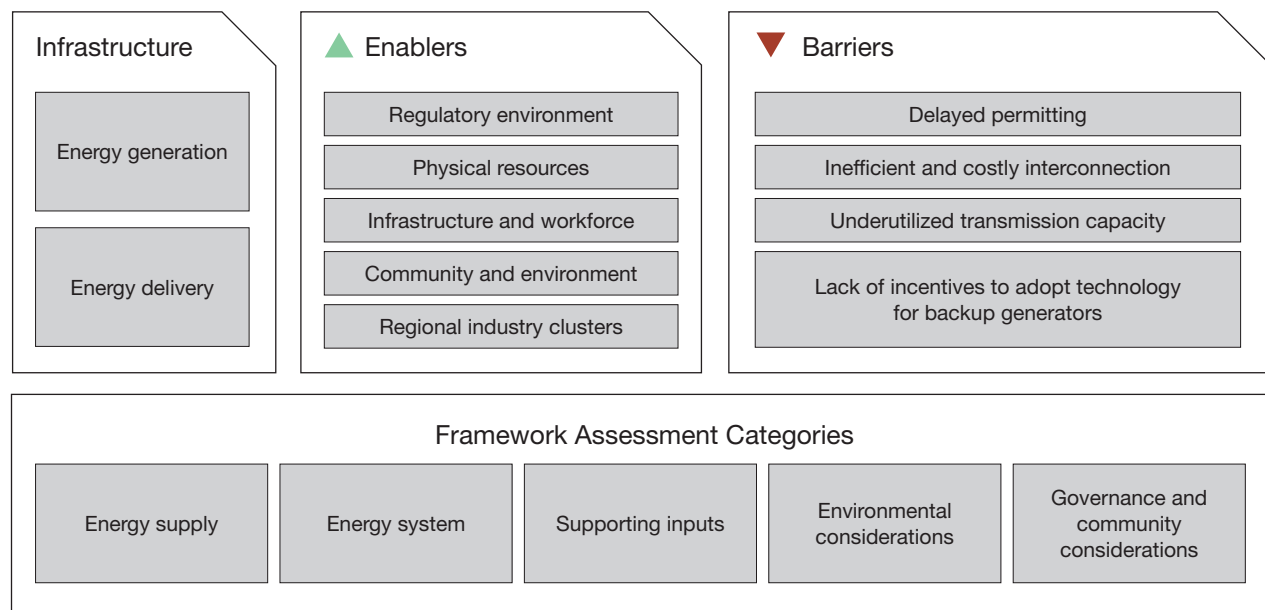
Many sites fell into the moderate-potential tier. These locations often had promising characteristics but also faced one or more constraints that could limit development unless addressed through targeted action. Examples included **Colossus**, **Oak Ridge National Laboratory**, **Los Alamos National Laboratory**, and retired or retiring plant sites, such as **Plant Wansley** and the **Cumberland Fossil Plant**.

Low-potential sites often faced multiple constraints at once, especially around grid conditions, environmental risks, or governance and community factors. More generally, the analysis suggests that site selection for AI infrastructure is best understood as a process of managing trade-offs rather than finding a perfect location.

TABLE 1
Summary of Assessment Categories

Category	What It Captures	Why It Matters for Energy-Related Site Suitability
Energy supply	On-site and nearby generation potential, resource diversity, and existing transmission and distribution (T&D) access	Helps indicate how much power may be available locally or can be delivered to the site
Energy system	Grid capacity additions, reliability, interconnection timelines, and planning environment	Shapes whether power can be connected in time and with sufficient reliability
Supporting inputs	Land, water, workforce, construction costs, and enabling conditions relevant to large energy-intensive development	Affects how quickly and feasibly energy infrastructure and data center capacity can be developed
Environmental considerations	Cooling needs, air quality constraints, and natural disaster risk	Influences operating efficiency, resilience, and the risk of delays or constraints on energy development
Governance and community considerations	Regulatory environment, affordability concerns, and local opposition that could affect infrastructure development	Affects whether energy and data center projects can secure approval, public support, and long-term operating viability

FIGURE 1
Conceptual Framework for Artificial Intelligence Data Center Evaluation



NOTE: This framework consists of the desirable site characteristics, as can be seen in terms of existing infrastructure, enablers, and barriers. The assessment involved a multicriteria decisionmaking process to assess potential sites against five categories that encompass infrastructure, enablers, and barriers. Sites that score well on the framework should have access to multiple enablers and encounter minimal barriers.

Applying the framework across the study sites showed that relative site suitability varies considerably and depends on trade-offs across energy, infrastructure, environmental, and governance factors.

Table 2 summarizes how the assessed sites were grouped into high-, moderate-, and low-potential tiers and highlights the trade-offs across the five energy-related assessment categories.

Existing Infrastructure Offers a Major Advantage

A central finding of the research is that using existing infrastructure offers a substantial advantage for meeting the 2030 timeline. Building major new T&D infrastructure is difficult to complete on that schedule because of permitting requirements, regulatory hurdles, supply chain challenges, and construction delays.

This makes previously developed sites especially attractive. Former industrial campuses, federal facilities, and retired or retiring power plant sites can offer existing substations, transmission connections, and local distribution infrastructure that reduce the time and cost needed to support large new loads. In some

cases, these sites may also have cleared land, industrial zoning, and legacy interconnection arrangements that make them more viable than undeveloped alternatives.

How Much Power Could One Site Support by 2030?

To estimate the maximum amount of power that might realistically be available at a single site by 2030, the researchers assessed three retired or retiring power plant locations. These sites were selected because they already benefit from substantial T&D infrastructure and offer opportunities for repowering or reuse.

Among them, **Rockport Power Plant in Indiana** had the highest estimated potential, at **about 4.2 gigawatts**. This estimate reflects both the site's strong existing transmission infrastructure and the possibility of using or repurposing power generation assets at the location.

Given current constraints on building major T&D infrastructure before 2030, this estimate likely represents the practical upper bound of grid-connected power that could be made available at a single U.S. site on that timeline. This finding suggests that the U.S.

TABLE 2

Site Assessment Summary, by Level of Energy Potential and Site Identifier

Site Identifier	Site ^a	Energy Supply	Energy System	Supporting Inputs	Environmental Considerations	Governance and Community Considerations
High potential						
1	Stargate Project ^b	B	B	A	C	B
18	Pantex Plant	B	C	B	C	A
19	Kansas City National Security Campus	A	C	C	C	C
21	Rockport Power Plant	B	C	B	B	B
Moderate potential						
2	Colossus	B	C	B	C	B
3	Idaho National Laboratory	C	C	C	B	C
4	Paducah Gaseous Diffusion Plant	C	C	C	C	C
5	Portsmouth Gaseous Diffusion Plant	C	C	B	B	C
9	National Energy Technology Laboratory Morgantown	C	D	B	B	C
12	Oak Ridge National Laboratory	B	C	C	B	B
13	Pacific Northwest National Laboratory	C	C	C	B	B
15	Los Alamos National Laboratory	C	B	B	B	C
16	Sandia National Laboratories	C	C	B	B	C
17	Savannah River Site	C	C	C	C	C
20	Cumberland Fossil Plant	C	C	C	B	C
22	Plant Wansley	B	C	B	C	B
Low potential						
6	Argonne National Laboratory	C	D	C	C	D
7	Brookhaven National Laboratory	C	C	C	D	D
8	Fermi National Accelerator Laboratory	C	D	C	C	B
10	National Energy Technology Laboratory Pittsburgh	D	D	B	D	C
11	National Laboratory of the Rockies	D	C	C	D	B
14	Princeton Plasma Physics Laboratory	C	D	C	D	D

NOTE: The letter grades indicate approximate potential energy capacity available at that site: A = no constraints; B = high potential; C = moderate potential; and D = low potential. See the accompanying research report, *Evaluating Potential Artificial Intelligence Energy Capacity at Different Data Center Sites*, for a full description of the various indicators, criteria, and category grades based on the framework used.

^a Site identifiers are as follows: 1–2 are industry-identified sites, 3–19 are DOE-identified sites, and 20–22 are retired or retiring power plant locations assessed separately.

^b The project is also known by its development code name, “Project Ludicrous,” and its location, Lancium Clean Campus.

power system could struggle to support the largest projected centralized frontier AI data centers without major infrastructure reuse, significant on-site power generation, or a more distributed compute strategy.

Policy Implications

Expanding AI data center capacity by 2030 will require more than identifying available land and power in the abstract. It will require focusing on the specific conditions that make some sites more feasible than others. RAND researchers suggest several priorities:

- **Target infrastructure improvements and investments.** Address site-level bottlenecks, such as a lack of substations, T&D upgrades, or interconnection constraints.

- **Prioritize sites with reusable infrastructure.** Especially important are industrial campuses, former manufacturing facilities, and retired or retiring power plants.
- **Account for environmental and reliability risks early.** These include water stress, natural disaster exposure, and cooling burdens.
- **Build community buy-in.** Early and transparent engagement can reduce opposition, litigation risk, and permitting delays.
- **Use an energy screening framework, such as the one presented in this brief, for site selection.** Policymakers and developers can apply the framework to candidate sites, preserve favorable site characteristics, and focus remediation efforts on addressable weaknesses.

This research brief was written by Ismael Arciniegas Rueda and Robin Wang and describes work done in RAND Global and Emerging Risks and documented in *Evaluating Potential Artificial Intelligence Energy Capacity at Different Data Center Sites*, by Ismael Arciniegas Rueda, Austin Smidt, Robin Wang, Hye Min Park, David Gill, and Henri van Soest, RR-A3845-3, 2026 (available at www.rand.org/t/RR-A3845-3). To view this research brief online, visit www.rand.org/t/RBA3845-3. RAND is a research organization that develops solutions to public policy challenges to help make communities throughout the world safer and more secure, healthier and more prosperous. RAND is nonprofit, nonpartisan, and committed to the public interest. RAND's publications do not necessarily reflect the opinions of its research clients and sponsors. **RAND**® is a registered trademark.

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